

**Course Title:** Peripherals, Interfacing and Embedded Systems Lab (CSE-4640)

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**Lab # 2**

*Interfacing with LEDs Lights through 8255A in MDA 8086 Kit.*

**Objective:**

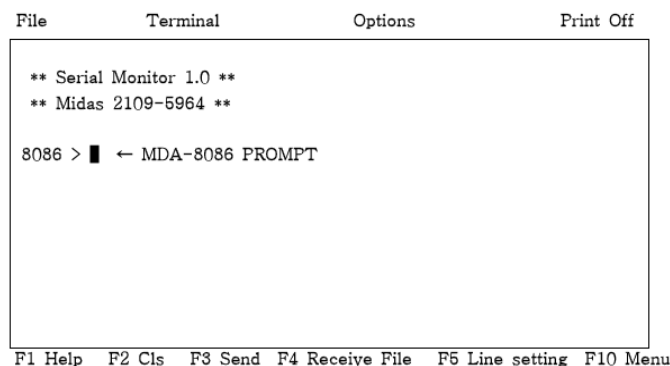
To understand the basic interfacing concept through implementing a model for with LEDs through 8255A interfacing of MDA 8086 trainer Kit.

**Theory:**

- **Serial Monitor Mode**

Serial monitor is the basic monitor program mode to have data communication between MDA-8086 and Computer (i.e., PC). To use serial monitor mode, move jumper P1 which located on the PCB like the following figure.

To connect the serial interface of the MDA-8086 with the Computer (PC) interface and run the WinComm program in PC to have the following screen shot after pressing RES key.



- **Operation Serial Monitor Command**

User can only use command which stored at serial monitor using WinComm. Serial monitor can execute to command when user type the command and then press ENTER key. If there is no any command at serial monitor, error message will be displayed with bell sound and serial monitor prompt will be displayed again.

8086 >?

**HELP Command**

**E** segment : offset.....: Enter Data To Memory

**D** segment : offset length.....: Dump Memory Contents

**R** [register name].....: Register Display & Change

**M** address1, length, address2.....: Move Memory From 1 to 2

**F** address, length, data.....: Fill Memory With Any Data

**L** Return key.....: Program Down Load

**G** segment : offset.....: Execute Program

**T**.....: Program 1 step execute

- **Basic Command Syntax:**

## 1. Memory Modify Command

**Syntax:** E segment: offset

**Purpose:** This command is used to enter data to memory.

**Example:**

```
8086 > E 0000:1000
      0000:1000 FF ? 11
      0000:1001 FF ? 22
      0000:1002 FF ? 33
      0000:1003 FF ? 44
      0000:1004 FF ? 55
      0000:1005 FF ? / (Offset decrement)
      0000:1004 55 ?
```

## 2. Memory Display Command

**Syntax:** D segment: offset

**Purpose:** This command is used to display the data stored in memory.

**Example:**

```
8086> D 0000:1000
0000:1000 11 22 33 44 55 FF FF FF - FF FF FF FF FF FF FF FF
0000:1020 FF FF FF FF FF FF FF FF - FF FF FF FF FF FF FF FF
```

## 3. Display Register Command

**Syntax:** R

**Purpose:** The R command is used to display the 8086 processor registers

**Example:**

```
8086 > R

AX=0000 BX=0000 CX=0000 DX=0000
SP=0540 BP=0000 SI=0000 DI=0000
DS=0000 ES=0000 SS=0000 CS=0000
IP=1000 FL=0000 = . . . . .
```

**To change individual register:-**

```
8086 > R AX
```

```
AX = 0000 1234
```

```
8086 > R BX
```

```
BX = 0000 4567
```

```
8086 > R CX
```

```
CX = 0000 7788
```

```
8086 > R DX
```

```
DX = 0000 1111
```

## 4. Program Download and Execute Command

The L command moves object data in hexa format from an external devices to memory.

8086 >L

Down load start !! ← ( Note : See section 5. Serial monitor experiment)

:14100000B83412BB7856B90010BA00208BF08BFBD0030BC08

:0910140000408EDA8ED18EC0CCB2

:00

OK Completed !!

☞ Set IP

8086 >R IP

IP=1000

8086 >T

AX=1234 BX=4567 CX=7788 DX=1111

SP=0540 BP=0000 SI=0000 DI=0000

DS=0000 ES=0000 SS=0000 CS=0000

IP=1003 FL=0100 = . . . t . . . .

↓

Next address

☞ Execute program command

Segment Offset

↓

↓

8086 >G 0000:1000

Execute Address = 0000:1000

### • LEDs in MDA 8086 Kit

There are 4 LEDs namely RED (L1), GREEN (L2), YELLOW (L3) and RED (L4) inside the MDA-8086 trainer kit and can be modeled to design a simpler application. This requires 8255 PPI ports which are already connected to the 4 LEDs internally. Through a code we can access these ports and provide binary or hex values to switch on the required LED (on or off). In order to turn a particular LED ON, a logical '1' should be provided to a particular port. Note that only Port B of 8255A PPI is used in the following example code to control the LEDs.

|                     |                    |
|---------------------|--------------------|
| RED<br><b>L1</b>    | GREEN<br><b>L2</b> |
| YELLOW<br><b>L3</b> | RED<br><b>L4</b>   |

Different ports of Programmable Peripheral Interface (PPI) 8255A is used for switching on LEDs, the address of the ports are:

Port A: 19h

Port C: 1Dh

Port B: 1Bh

Control Register: 1FH

To control the LEDs, Port B will be used for the value to select the LEDs and Port A and C will be set with constant values. In-case of Port B, pass a value of '11110001' to select the L1 LED and pass a value of '11110010' to select the L2 LED and so on.

Setup jumper cap of P2 as following:



- **Example:** Example code to illuminate LEDs with a sequence of L1, L2, L3 and L4.

```

CODE SEGMENT
    ASSUME CS:CODE, DS:CODE, ES:CODE, SS:CODE
PPIC_C EQU 1FH          ; Control register address
PPIC EQU 1DH            ; Port C address
PPIB EQU 1BH            ; Port B address
PPIA EQU 19H            ; Port A address

    ORG 1000H
    MOV AL, 10000000B    ; Control register initialization with a 8255 Mode set
    OUT PPIC_C, AL
    MOV AL, 11111111B
    OUT PPIA, AL
    MOV AL, 00000000B
    OUT PPIC, AL
L1:  MOV AL, 11110001B
L2:  OUT PPIB, AL        ; Select a L1 LED with AL value and make it ON
    CALL TIMER
    SHL AL, 1
    TEST AL, 00010000B   ; Perform Logical AND and set Zero Flag (ZF), SF, PF
    JNZ L1
    OR AL, 11110000B     ; Perform Logical OR and store the result in AL
    JMP L2
    INT 3                ; Single-step interrupt
TIMER: MOV CX, 1
TIMER2: PUSH CX
        MOV CX, 0
TIMER1: NOP
        NOP
        NOP
        NOP
        LOOP TIMER1
        POP CX
        LOOP TIMER2
        RET

CODE ENDS
END

```

### Tasks to do:

1. Write a program to illuminate LEDs for the implementation of Simple Traffic Control Light Signaling model. Vehicles will be in STOP (with red color), WAIT (with yellow color) & GO (i.e., with green color) states for a moderately long time and vice-versa:

